METRIC
MIL-PRF-24596C(SH)
w/AMENDMENT 1
12 December 2025
SUPERSEDING
MIL-PRF-24596C(SH)
4 January 2022

PERFORMANCE SPECIFICATION

COATING COMPOUNDS, NONFLAMING, FIRE-RESISTANT

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers nonflaming, fire-resistant, non-intumescent coatings that are used where volatile organic compound (VOC) content air pollution regulations apply for solvents in paints and for application over selected surfaces.

1.2 Classification. Coating compounds covered by this specification are of the following classes, grades, and applications, as specified (see 6.2).

1.2.1 Classes. The classes of coating compounds are as follows:

a. Class 1 - Surface ship, general usage.

b. Class 2 - Submarine, general usage.

c. Class 4 - DELETED

d. Class 5 - Navy formula 25A.

1.2.2 Grades. The grades of coating compounds are as follows:

a. Grade A - Water-based, air dried at temperatures less than 38 °C (100 °F); VOC not to exceed 175 grams per liter (g/L) (1.46 pounds/gallon) of paint.

b. Grade B - Water-based, requiring baking at temperatures above 87 °C (188 °F); VOC not to exceed 250 g/L (2.1 pounds/gallon) of paint.

c. Grade C - Solvent-based, air dried at temperatures less than 38 °C (100 °F); VOC not to exceed 250 g/L (2.1 pounds/gallon) of paint.

d. Grade E - Solvent-based, requiring baking at temperatures above 87 °C (188 °F); VOC not to exceed 250 g/L (2.1 pounds/gallon) of paint.

1.2.3 Applications. The applications of coating compounds are as follows:

a. Application A - Steel.

b. Application B - Aluminum.

c. Application C - Glass-reinforced plastics (GRP).

d. Application D - Wood.

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 8010

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e. Application E - Elastomeric foam insulation.

1.3 Part or identifying number (PIN). PINs to be used for nonflaming and fire-protective coatings acquired to this specification are created as follows:

M	**24596**	-	**X**	**X**	**X**
Prefix for military specification	Specification number		Class (see code below)	Grade (see code below)	Application (see code below)

Class code		**Grade code**		**Application code**	
Class	Code	Grade	Code	Application	Code
1	A	A	1	A	A
2	B	B	2	B	B
5	C	C	3	C	C
		E	4	D	D
				E	E

Examples:

a. M24596-A2C
b. M24596-C4B

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

DELETED

TT-P-645 - Paint, Alkyd Type, Formula Number 84

FEDERAL STANDARDS

FED-STD-141 - Paint, Varnish, Lacquer and Related Materials: Methods of Inspection, Sampling and Testing

DEPARTMENT OF DEFENSE SPECIFICATIONS

DELETED

MIL-DTL-24441/20 - Paint, Epoxy-Polyamide, Green Primer, Formula 150, Type III

MIL-DTL-24441/29 - Paint, Epoxy-Polyamide, Green Primer, Formula 150, Type IV

MIL-DTL-24607 - Enamel, Interior, Nonflaming (Dry), Chlorinated Alkyd Resin

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS (CFR)

29 CFR 1910	-	Occupational Safety and Health Standards
29 CFR 1910.1000, Subpart Z	-	Toxic and Hazardous Substances
29 CFR 1910.1200	-	Hazard Communication
29 CFR 1990	-	Identification, Classification, and Regulation of Potential Occupational Carcinogens
40 CFR Chapter 1, Part 60, Appendix A, Method 24	-	Determination of Volatile Matter Content, Water Content, Density, Volume Solids, and Weight Solids of Surface Coatings
40 CFR 63, Subpart II	-	National Emission Standards for Shipbuilding and Ship Repair (Surface Coating)
40 CFR 63, Appendix A, Method 311	-	Method 311 – Analysis of Hazardous Air Pollutant Compounds in Paints and Coatings by Direct Injection Into a Gas Chromatograph

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40 CFR 82	-	Protection of Stratospheric Ozone
40 CFR 261.24	-	Toxicity Characteristic
40 CFR 355, Appendixes A and B	-	The List of Extremely Hazardous Substances and Their Threshold Planning Quantities
40 CFR 372.65, Subpart D	-	Specific Toxic Chemical Listings
49 CFR 171-178	-	Hazardous Materials Regulations

(Copies of these documents are available online at www.ecfr.gov.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9510-AB-ATM-010 - Nuclear Powered Submarine Atmosphere Control Manual; Volume 1

(Copies of the chapter titled "Material Control Program" are available by email request to CommandStandards@navy.mil.)

T9070-AL-DPC-020/077-2 - NAVSEA Hazardous Material Avoidance Process

(Copies of this document are available online via Model Based Product Support (MBPS) at https://mbps.navsea.cloud.navy.mil/Windchill. To request an MBPS account, submit a ticket to the ServiceNow Assistance Portal (SNAP), https://snap.navair.navy.mil/mbps (DoD PKI Certificate required).)

OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

ID-191 - Polarized Light Microscopy of Asbestos

(Copies of this document are available online at www.osha.gov.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASSOCIATION FOR MATERIALS PROTECTION AND PERFORMANCE (AMPP)

SSPC-SP 10/NACE No. 2 - Near-White Metal Blast Cleaning

(Copies of this document are available online at www.ampp.org.)

ASTM INTERNATIONAL

ASTM B117	-	Standard Practice for Operating Salt Spray (Fog) Apparatus
ASTM D93	-	Standard Test Methods for Flash Point by Pensky-Martens Closed Cup Tester
ASTM D522/D522M	-	Standard Test Methods for Mandrel Bend Test of Attached Organic Coatings
ASTM D523	-	Standard Test Method for Specular Gloss
ASTM D562	-	Standard Test Method for Consistency of Paints Measuring Krebs Unit (KU) Viscosity Using a Stormer-Type Viscometer
ASTM D609	-	Standard Practice for Preparation of Cold-Rolled Steel Panels for Testing Paint, Varnish, Conversion Coatings, and Related Coating Products
ASTM D610	-	Standard Practice for Evaluating Degree of Rusting on Painted Steel Surfaces

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ASTM D714	-	Standard Test Method for Evaluating Degree of Blistering of Paints
ASTM D1308	-	Standard Test Method for Effect of Household Chemicals on Clear and Pigmented Organic Finishes
ASTM D1475	-	Standard Test Method for Density of Liquid Coatings, Inks, and Related Products
ASTM D1640/D1640M	-	Standard Test Methods for Drying, Curing, or Film Formation of Organic Coatings
ASTM D1729	-	Standard Practice for Visual Appraisal of Colors and Color Differences of Diffusely-Illuminated Opaque Materials
ASTM D1849	-	Standard Test Method for Package Stability of Paint
ASTM D2243	-	Standard Test Method for Freeze-Thaw Resistance of Water-Borne Coatings
ASTM D2247	-	Standard Practice for Testing Water Resistance of Coatings in 100% Relative Humidity
ASTM D2621	-	Standard Test Method for Infrared Identification of Vehicle Solids from Solvent-Reducible Paints
ASTM D2698	-	Standard Test Method for Determination of the Pigment Content of Solvent-Reducible Paints by High-Speed Centrifuging
ASTM D2805	-	Standard Test Method for Hiding Power of Paints by Reflectometry
ASTM D3273	-	Standard Test Method for Resistance to Growth of Mold on the Surface of Interior Coatings in an Environmental Chamber
ASTM D3274	-	Standard Test Method for Evaluating Degree of Surface Disfigurement of Paint Films by Fungal or Algal Growth, or Soil and Dirt Accumulation
ASTM D3278	-	Standard Test Methods for Flash Point of Liquids by Small Scale Closed-Cup Apparatus
ASTM D3723	-	Standard Test Method for Pigment Content of Water-Emulsion Paints by Low-Temperature Ashing
ASTM D3828	-	Standard Test Methods for Flash Point by Small Scale Closed Cup Tester
ASTM D4541	-	Standard Test Method for Pull-Off Strength of Coatings Using Portable Adhesion Testers
ASTM D4828	-	Standard Test Methods for Practical Washability of Organic Coatings
ASTM D5590	-	Standard Test Method for Determining the Resistance of Paint Films and Related Coatings to Fungal Defacement by Accelerated Four-Week Agar Plate Assay
ASTM D6677	-	Standard Test Method for Evaluating Adhesion by Knife
ASTM E162	-	Standard Test Method for Surface Flammability of Materials Using a Radiant Heat Energy Source
ASTM E313	-	Standard Practice for Calculating Yellowness and Whiteness Indices from Instrumentally Measured Color Coordinates

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ASTM E662	-	Standard Test Method for Specific Optical Density of Smoke Generated by Solid Materials
ASTM E800	-	Standard Guide for Measurement of Gases Present or Generated During Fires
ASTM E1252	-	Standard Practice for General Techniques for Obtaining Infrared Spectra for Qualitative Analysis
ASTM F683	-	Standard Practice for Selection and Application of Thermal Insulation for Piping and Machinery
ASTM F718	-	Standard Specification for Shipbuilders and Marine Paints and Coatings Product/Procedure Data Sheet

(Copies of these documents are online at www.astm.org.)

INTERNATIONAL AGENCY FOR RESEARCH ON CANCER (IARC)

IARC Monographs on the Evaluation of Carcinogenic Risks to Humans

(Copies of this document are available online at www.who.int/bookorders.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO/IEC 17025	-	General Requirements for the Competence of Testing and Calibration Laboratories

(Copies of this document are available online at www.iso.org/home.html.)

SAE INTERNATIONAL

SAE AMS-STD-595	-	Colors Used in Government Procurement
SAE AMS-STD-595/22563	-	Orange, Semigloss (Beachwood)
SAE AMS-STD-595/24585	-	Green, Semigloss (Postal Green)
SAE AMS-STD-595/25526	-	Blue, Semigloss (Pastel Blue)
SAE AMS-STD-595/26307	-	Gray, Semigloss (Bulkhead Machinery Gray #30)
SAE AMS-STD-595/35044	-	Blue, Flat (Aircraft Insignia Blue, ANA 605)
SAE AMS-STD-595/27925	-	Misc, Semigloss
SAE AMS-STD-595/27880	-	Misc, Semigloss

(Copies of these documents are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Coating compounds furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

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3.1.1 Qualification of commercial products specified as Navy formula 25A. Commercial products certified to meet the requirements of Navy formula 25A by the manufacturer in the following colors, SAE AMS-STD-595 colors, SAE AMS-STD-595/24585 (postal green), SAE AMS-STD-595/26307 (bulkhead machinery gray #30), and SAE AMS-STD-595/27880 (misc, semigloss), shall be accepted as meeting the requirements of class 1, grade A, all applications. These accepted products shall be qualified without testing to the class 1, grade A requirements herein. See appendix B.

3.2 Composition. Commercial products shall be certified to meet the requirements of Navy formula 25A (see appendix B) by the manufacturer. Coatings shall be furnished in accordance with the requirements of the class, grade, and application specified (see 6.2). The composition of the coating furnished to this specification shall be the responsibility of the manufacturer, except as otherwise limited by this specification or 29 CFR 1910.1000, 29 CFR 1990, 49 CFR 171-178, 40 CFR 355, appendices A and B, 40 CFR 372.65, Subpart D, 40 CFR 82, and International Agency for Research on Cancer (IARC) Monographs (see 3.10).

3.3 Color.

3.3.1 Coating colors. Coatings shall be qualified color numbers (see 4.3 for extension of qualification to other colors). Colors not listed in table I shall be qualified individually.

TABLE I. Historical color names.

SAE AMS-STD-595A color chip number	Historical coating name
27880 – Misc, Semigloss	Soft White
27925 – Misc, Semigloss	Bright White
24585 – Postal Green	Pastel Green
35044 – Aircraft Insignia Blue, ANA 605	Insignia Blue
26307 – Bulkhead Machinery Gray #30	Bulkhead Gray
25526 – Pastel Blue	Pastel Blue
22563 – Beachwood	Beach Sand

3.3.2 Ready-mixed. Colors shall be supplied as ready-mixed products.

3.4 Identification characteristics. Coatings shall be identified by product name, color, and a unique alpha, numeric, or alpha-numeric identifier. Values for identification (ID) characteristics and the mixing, application instructions, and test methodology used to determine those characteristics indicated by an “X” in table II shall be provided by manufacturers seeking qualification (see 6.2). The values shall be established for each coating system prior to qualification testing. The purpose of these values is to serve as a basis for determining that the material being offered is essentially the same as that which was approved under qualification testing. Identification characteristics for all classes, grades, and compositions shall be conducted at 16 to 27 °C (60 to 80 °F) and 50±10 percent relative humidity. Subject to review by the Naval Sea Systems Command (NAVSEA), alternative identification characteristics (for example, infrared spectrographic examination and chemical analysis) may be substituted for the characteristics in table II. Manufacturers shall provide ASTM method(s), FED-STD-141 methods, other consensus standards, or a copy of unique test methodology and any other necessary information on methodology used to determine reported ID characteristics, including those in table II.

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TABLE II. Required identification characteristics.

Characteristic	Single component materials
Chemical nature of resin(s)	X
Percent of principal constituents (5 percent by weight [wt%] or more of total)	X
Pigment, wt%	X
Nonvolatile vehicle, wt%	X
Total solids, wt%	X
Color (as descriptive name and SAE AMS-STD-595 number)	X
Volatile, wt%	X
Density, grams/liter (pounds/gallon)	X
Viscosity, Kreb units (KU) or Poise	X
Flash point as Setaflash, or Pensky-Martin closed cup, or equal, °C	X
Drying time to touch and dry hard time or curing time, hours	X
Specular gloss, gloss units (GU)	X
Hiding power, (contrast ratio)	X
Volatile organic content, grams/liter (pounds/gallon)	X
Mixing instructions	X
Pot life	X
Shelf life	X

3.5 Quantitative requirements. Unless otherwise stated in this section, the following requirements apply to all classes, grades, and applications.

3.5.1 VOC. When tested in accordance with 4.6.2, the VOC of appropriate grades shall not exceed the value listed:

Grade	Maximum VOC	
	Grams per liter	Pounds per gallon
A	175	1.46
B	250	2.08
C	250	2.08
E	250	2.08

3.5.2 Volatile organic hazardous air pollutant (VOHAP) content. When evaluated as specified in 4.6.3, the total VOHAP content, as defined by the 40 CFR 63, Subpart II (National Emission Standards for Shipbuilding and Ship Repair (Surface Coating)) requirements for shipbuilding and ship repair, of each system component, shall not exceed the VOC limit prescribed for the specific types (see 3.5.1).

3.5.3 Hazardous pigments and additives. When tested in accordance with 4.6.4, the content of each soluble metal and the total content of each metal of the coating shall be less than the values listed in table III. In addition, crystalline silica, asbestos, and asbestos from pigment shall not exceed 0.1 percent by weight.

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TABLE III. Metals content of coatings.

Metal and its compound in each coating dry film	Soluble metal, maximum, mg/L	Total metals content, maximum, wt%
Antimony	15.0	0.015
Arsenic	5.0	0.005
Barium (excluding barite)	100.0	0.10
Beryllium	0.75	0.0002
Cadmium	1.0	0.0005
Chromium(VI) compounds	1.0	0.001
Chromium, including chromium(III)	560.0	0.56
Cobalt 1/	50.0	0.005
Copper	25.0	0.01
Fluoride salts	180.0	0.18
Lead	5.0	0.005
Mercury	0.2	0.0002
Molybdenum	350.0	0.35
Nickel	20.0	0.02
Selenium	1.0	0.002
Silver	5.0	0.001
Tantalum	100.0	0.100
Thallium	7.0	0.007
Tungsten	100.0	0.100
Vanadium	24.0	0.01
Zinc	250.0	0.25

NOTE:

1/ Total cobalt content may exceed 0.005 wt% (up to 0.2 wt%) only if a cobalt drier is used to effect proper drying. However, soluble cobalt content may not exceed table III requirements.

3.5.4 Flash point. When tested in accordance with table IV, the flash point of the coating shall be not less than 38 °C (100 °F).

3.5.5 Gloss. When tested in accordance with 4.6.1, the 60-degree specular gloss shall be between 45 and 60 GU with the exception of color number 35044 in which the acceptable range shall be less than 12 GU.

3.5.6 Yellowness. When tested in accordance with 4.6.5, the yellowness index of panels in light or dark storage shall not increase more than 3 units.

3.6 Qualitative requirements. Unless otherwise stated in this section, the following requirements apply to all classes, grades, and applications.

3.6.1 Condition in container. When tested in accordance with table IV, a freshly opened full container of the enamel shall be free of grit, seeds, skins, lumps, putrefaction, rancidity, or livering, and shall show no more surface float (separated pigments or other ingredients on the surface of the liquid enamel), pigment settling, or caking than can be readily re-incorporated to a smooth, uniform state. The container shall be free of corrosion.

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3.6.2 Mixing properties. When tested in accordance with 4.6.6, the coating shall be free from grit, seeds, skins, or lumps, and shall result in a smooth, uniform mixture. After aging, the coating shall show no signs of gelation or separation.

3.6.3 Spraying properties. When tested in accordance with 4.6.7, the coating shall spray satisfactorily and shall show no running, sagging, or streaking. The dried film shall show no dusting, mottling, or color separation and shall present a smooth finish.

3.6.4 Brushing properties. When tested in accordance with 4.6.8, the coating shall brush satisfactorily and dry to a smooth, uniform film that is free from seeds, runs, or sags.

3.6.5 Flexibility. When tested in accordance with 4.6.9, a film of coating shall show no checking, cracking, or flaking.

3.6.6 Knife test adhesion. When tested in accordance with 4.6.10, the coating film shall be difficult to furrow off with a knife, shall not flake, chip, or powder, and shall show neither loss of inter-coat adhesion nor loss of adhesion to the surface of the substrate. The knife cut shall show beveled edges.

3.6.6.1 Adhesion. When tested in accordance with table IV, a film of coating shall adhere tightly to the test panel.

3.6.7 Fluid resistance.

3.6.7.1 Water resistance (does not apply to grade A and B water-based coatings). When tested in accordance with 4.6.11, a film of coating shall show no wrinkling or blistering immediately after removal of the wetted sponges. The coating shall be no more than slightly affected when examined 2 hours after removal of the wetted sponges. After 24 hours of air drying, the portion of the panel that was covered by the wetted sponges shall be almost indistinguishable with regard to hardness and adhesion from the portion that was not immersed. A slight discoloration of the portion of the panel covered by the wetted sponges shall be permitted.

3.6.7.2 Hydrocarbon fluid resistance (does not apply to grade A and B water-based coatings). When tested in accordance with 4.6.12, a film of coating shall show no blistering or wrinkling and no more than a slight whitening or softening upon removal of the fluid wetted sponge. After 2 hours of air drying, the portion of the panel that was covered by the wetted sponge shall be almost indistinguishable with regard to hardness, color, and gloss from a panel prepared at the same time, but not immersed.

3.6.7.3 Salt spray resistance (does not apply to application C, D, and E coatings). When tested in accordance with 4.6.13, a film of coating examined immediately after removal from the test shall show no more than a trace of corrosion in accordance with ASTM D610 and not more than five scattered blisters no larger than 1 millimeter (0.04 inch) in diameter.

3.6.7.4 Condensation blister resistance. When tested in accordance with 4.6.14, a film of coating shall show a maximum ASTM D714 blister size rating of “8” and a frequency rating of “few”. Blisters within 12 millimeters (0.47 inch) of all edges shall be disregarded.

3.6.8 Resistance to exudate formation (grade A and B water-based coatings only). When tested in accordance with 4.6.15, a film of coating shall exude no fluid (liquid) material from the paint film and the paint film shall not be tacky.

3.6.9 Washability. When tested in accordance with 4.6.16, a film of coating shall retain a minimum of 50 percent of its original gloss.

3.6.10 Flash rust resistance (grade A and B, application A water-based coatings only). When tested in accordance with 4.6.17, there shall be no evidence of flash rust or pinhole corrosion.

3.6.11 Resistance to growth of mold.

3.6.11.1 Standard mold resistance. When tested in accordance with 4.6.18.1, after 4 weeks, the test panels shall have a minimum ASTM D3274 rating of “8”.

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3.6.11.2 High mold resistance (class 1, grade A coatings only). When tested in accordance with 4.6.18.2, after 6 weeks, the test panels shall have an ASTM D5590 rating of "0" and a zone of inhibition (ZOI) around the sample of 1 millimeter (0.04 inch) minimum.

3.6.12 Freeze-thaw resistance (grades A and B only). When tested for five cycles in accordance with 4.6.19, the coating shall not exhibit broken emulsion or be agglomerated or coagulated. There shall be no change in hiding power, gloss, fineness of grind, or speckiness. The color shall match the standard color chip of the appropriate color. Viscosity shall be within 10 percent of the viscosity of the control. Brushability and sprayability shall be the same as the control.

3.6.13 Recoatability. When tested in accordance with 4.6.20, the paint shall not wrinkle, blister, or lift. When cut with a knife blade, the overcoat paint shall not lose adhesion or flake away from the test paint film. The surface of the recoat paint shall be smooth.

3.7 Thermal properties. Unless otherwise specified (see 6.2), all paints regardless of application shall conform to the requirements in 3.7.1 through 3.7.4.

3.7.1 Radiant panel. When tested in accordance with 4.6.21.2, coating systems shall have the following maximum flame spread indices:

Application	Maximum flame spread index
A	5
B	5
C	25
D	25
E	50

3.7.2 Smoke density. When tested in accordance with 4.6.21.3, coating systems shall have the following maximum smoke generation values:

Application	Maximum allowed smoke density
A	15
B	15
C	20
D	30
E	40

3.7.3 Resistance to ignition (application A only). When tested in accordance with 4.6.21.4, coatings shall not exhibit flaming, sparking, fumes or smoke, or burning drops in excess of that of any coating qualified to MIL-DTL-24607.

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3.7.4 Hazardous emissions. When tested in accordance with the requirements of 4.6.21.3, emissions of the indicated compounds shall not exceed the values listed:

Compound	Maximum value, parts per million (ppm)
HCN	30
HCl	30
CO	600

3.8 Condensation (class 4 only). DELETED

3.9 Label. A label instruction shall be prepared in accordance with 29 CFR 1910. Each container shall be affixed with a hazardous chemical warning label in accordance with 29 CFR 1910.1200. To comply with 40 CFR 63, Subpart II (National Emission Standard for Shipbuilding and Ship Repair (Surface Coatings)), the two following statements shall appear on each paint can label:

a. Certification that the paint in the container meets the 40 CFR 63, Subpart II requirements for shipbuilding and ship repair.

b. Statement of the ratio of volatile content to solids expressed as grams of volatile organic hazardous air pollutants (VOHAP) per liter of solids.

3.10 Toxicity (all classes, all grades, and all applications). When evaluated in accordance with 4.7, the coating system shall pose no serious or high risk to the health of personnel or the environment when used for its intended purpose (see 4.7 and 6.6).

3.11 Prohibited materials. The coating system shall not contain any chemicals categorized as “prohibited” in accordance with T9070-AL-DPC-020/077-2.

3.12 Off-gassing (class 2 only). The coating shall be evaluated for off-gassing in accordance with the requirements of 4.9. Based on the circumstances of use and the chemical nature of the coating, the Navy will determine whether off-gas testing is required or if an administrative assessment is acceptable. In order to be considered acceptable for use in submarines, the coating shall be assigned to either the “Permitted” or “Limited” category (see 4.9 and 6.7).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection tests shall consist of all tests specified in table IV and evaluation of paragraphs 4.7 through 4.10.

4.2.1 Changes to product. A change in material used in the manufacture of the coating system, which has been qualified, shall require written approval by the qualifying activity. Incorporation of any change that has not been approved by the qualifying activity shall require requalification of the item in question.

4.2.2 Navy formula 25A. Qualification shall be extended to certified coatings manufactured in accordance with the requirements of Navy formula 25A (see appendix B) for class 1, grades A and B, applications A, B, C, D, and E, and colors SAE AMS-STD-595/24585 (postal green), SAE AMS-STD-595/26307 (bulkhead machinery gray 30), and SAE AMS-STD-595/27880 (misc, semigloss) without testing. Other colors based on Navy formula 25A shall require qualifying activity approval in accordance with 4.2.1.

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4.3 Extension of qualification. Qualification to applications A (steel) and E (insulation) are suitable for qualification extension to applications B (aluminum), C (GRP), and D (wood). Qualification extension from application A to application E shall consist of knife and adhesion tests specified in table IV performed on elastomeric foam insulation samples prepared in accordance with 4.5.1.1.1. Qualification shall not be extended to paints in colors not listed in table I. Qualification without testing shall not be extended to Navy formula 25A in colors other than those listed in 4.2.2. Qualification to the colors of table I is extended for all paints provided the requirements of 3.3.1 are met. Any paints in colors not listed in table I shall be qualified separately as a commercial product.

4.4 Conformance inspection. Conformance inspection shall consist of all tests specified in table IV as conformance tests. At a minimum (see 6.2), conformance inspections shall be performed on the first lot and every 19,000 liters (5,000 gallons) thereafter. A lot shall consist of all components of a coating formulation from a single uniform batch or single uniform blend of batches offered for delivery at one time.

TABLE IV. Test procedures.

Item	Requirement	Test method	Conformance tests
Color	3.3, 3.4	4.6.1, ASTM D1729	X
Chemical nature of resin(s)	3.4	4.6.1, ASTM D2621, ASTM E1252	---
Principal constituents, %wt	3.4	4.6.1	---
Pigment, %wt	3.4	4.6.1, ASTM D2698, or ASTM D3723	---
Nonvolatile vehicle solids, %wt	3.4	4.6.1, ASTM D2698	---
Volatiles, %wt	3.4	4.6.1, ASTM D2698	---
Mass per unit volume	3.4	4.6.1, ASTM D1475	X
Viscosity	3.4	4.6.1, ASTM D562	X
Flash point	3.4, 3.5.4	4.6.1, ASTM D93, ASTM D3278, or ASTM D3828	X
Drying time	3.4	4.6.1, ASTM D1640/D1640M	X
Set to touch	3.4	4.6.1, ASTM D1640/D1640M	X
Dry hard	3.4	4.6.1, ASTM D1640/D1640M	---
Specular gloss	3.4, 3.5.5	4.6.1, ASTM D523	X
Hiding power (contrast ratio)	3.4	4.6.1, ASTM D2805	---
Adhesion	3.6.6.1	4.6.10, ASTM D4541	---
VOC	3.4, 3.5.1	4.6.2	---
VOHAP content	3.5.2	4.6.3	---
Hazardous pigments and additives	3.5.3	4.6.4	---
Yellowness	3.5.6	4.6.5, ASTM E313	---
Condition in container	3.6.1	4.4, ASTM D1849	---
Mixing properties	3.6.2	4.6.6	---
Spraying properties	3.6.3	4.6.7, FED-STD-141 method 4331.2	---
Brushing properties	3.6.4	4.6.8, FED-STD-141 method 4321.3	---

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TABLE IV. Test procedures – Continued.

Item	Requirement	Test method	Conformance tests
Flexibility	3.6.5	4.6.9, ASTM D522/D522M	---
Knife test	3.6.6	4.6.10, ASTM D6677	---
Resistance to water	3.6.7.1	4.6.11, ASTM D1308	---
Resistance to hydrocarbons	3.6.7.2	4.6.12, ASTM D1308	---
Resistance to salt spray	3.6.7.3	4.6.13, ASTM B117	---
Condensation blister resistance	3.6.7.4	4.6.14, ASTM D2247, ASTM D714	---
Resistance to exudates formation	3.6.8	4.6.15	---
Washability	3.6.9	4.6.16, ASTM D4828	---
Flash rust resistance	3.6.10	4.6.17	---
Standard mold resistance	3.6.11.1	4.6.18.1, ASTM D3273, ASTM D3274	---
High mold resistance	3.6.11.2	4.6.18.2, ASTM D5590	---
Freeze-thaw resistance	3.6.12	4.6.19, ASTM D2243	---
Recoatability	3.6.13	4.6.20	---
Radiant panel	3.7.1	4.6.21.2, ASTM E162	---
Smoke density	3.7.2	4.6.21.3, ASTM E662	---
Resistance to ignition	3.7.3	4.6.21.4	---
Hazardous emissions	3.7.4	4.6.21.3	---
Condensation	DELETED	DELETED	---
Toxicity	3.10	4.7, 40 CFR 261 part 24	---
Packaging	Contract	5.1	---
Label	3.9	4.11	---

4.5 Test conditions. Unless otherwise specified herein, all physical tests on the coating system shall be conducted at 16 to 27 °C (60 to 80 °F) and 50±10 percent relative humidity.

4.5.1 Test panels. Unless otherwise specified (see 6.2), test panels shall be prepared in accordance with 4.5.1.1 through 4.5.1.1.4.

4.5.1.1 General. Three sets of panels shall be prepared and all tests conducted in triplicates.

4.5.1.1.1 Test panels. Metal panels shall have nominal dimensions of 150- by 300- by 3- millimeters (6- by 12- by 0.125-inch) plate aluminum or cold rolled steel. GRP, wood, and elastomeric foam insulation in accordance with anti-sweat insulation of ASTM F683, Supplement S1.26 at a minimum thickness of 25 millimeters (1 inch) shall be adhered with the smooth side up to a metal support panel.

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4.5.1.1.2 Priming of test panels. One set of test panels shall be primed with paint in accordance with TT-P-645. The other set shall be primed with primer in accordance with MIL-DTL-24441/20 or MIL-DTL-24441/29. Only the face(s) of the panels subject to test exposure are required to be primed. Application D and E panels shall not be primed unless required for compatibility or adhesion such as water-based primer on application E materials. The dry film thickness of the primer shall be 0.042±0.012 millimeter (0.002±0.001 inch).

4.5.1.1.3 Conditioning of primed test panels. Unless otherwise specified (see 6.2), primed panels shall be dried (cured) for 1 week at normal laboratory conditions.

4.5.1.1.4 Application of test coatings. Unless otherwise specified (see 6.2), test coatings shall be applied in accordance with the manufacturer's data sheet or ASTM F718.

4.5.1.2 Surface preparation.

4.5.1.2.1 Application A (steel). Panels shall be mild steel. Test panels shall be degreased in accordance with method B, C, or D of ASTM D609. The degreased test panels shall be abrasive blasted to SSPC-SP 10/NACE No.2 near white metal with new, clean aluminum oxide to provide an average profile pattern of 0.051 to 0.076 millimeter (2 to 3 mils) on both sides. Dust shall be removed with clean, dry air or vacuumed away.

4.5.1.2.2 Application B (aluminum). Panels shall be aluminum. Test panels shall be degreased in accordance with method B, C, or D of ASTM D609. The degreased test panels shall be abrasive blasted with new, clean aluminum oxide to provide an average profile pattern of 0.051 to 0.076 millimeter (2 to 3 mils) on both sides. Dust shall be removed with clean, dry air or vacuumed away.

4.5.1.2.3 Application C (GRP). The face (smooth side) of panels shall be degreased and abraded with 100-grit abrasive paper or cloth (sandpaper) to a uniform appearance. Dust shall be removed with clean, dry air or vacuumed away.

4.5.1.2.4 Application D (wood). Panels shall be solid wood, not plywood. Panels shall be degreased and abraded with 100-grit abrasive paper or cloth (sandpaper) to a uniform appearance. Dust shall be removed with clean, dry air or vacuumed away.

4.5.1.2.5 Application E (anti-sweat insulation in accordance with ASTM F683, S1.26). The smooth side of the elastomeric foam shall be used for the test surface. Only the smooth side shall be prepared and painted. Panels shall be degreased. Dust shall be removed with clean, dry air or vacuumed away.

4.6 Test procedures. Tests shall be conducted in accordance with table IV.

4.6.1 Tests to verify ID characteristic values. Tests shall be conducted in accordance with the procedures of table IV. Results shall conform to the requirements of 3.4.

4.6.2 VOC. VOC shall be determined in accordance with 40 CFR Chapter 1, Subchapter C, Part 60, appendix A-7, method 24. Results shall conform to the requirements of 3.5.1.

4.6.3 VOHAP content. The VOHAP content of the coating shall be measured in accordance with 40 CFR 63, appendix A, method 311 (EPA Test method 311). Solvent fractions shall be identified in accordance with ASTM E1252 with the results recorded as percent weight of the total coating system. Alternate methods of analysis shall be approved by NAVSEA. When specified, formulation data may be used by manufacturers in lieu of testing to demonstrate compliance with 3.5.2. The formulation data shall have a consistent and quantitatively known relationship to the testing required. Calculation of individual VOHAP contents can be based on either manufacturer evaluation of batches or supplier data for raw materials used in the product. Results shall conform to the requirements of 3.5.2.

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4.6.4 Hazardous pigments and additives. Soluble and total metal content, except tantalum and tungsten, shall be determined on a dry paint film of the coatings in accordance with 40 CFR 261.24 (a), Toxicity Characteristic Leaching Procedure (TCLP). Asbestos shall be analyzed in accordance with OSHA method ID-191. Test results shall be reported as percent by weight of the dry enamel film. Soluble metal content shall be reported as milligrams per liter (mg/L). Total metal content shall be reported as percent weight of the dry enamel film (1 percent equaling 10,000 mg/L). The test results shall conform to the requirements of 3.5.3. Tantalum and tungsten soluble metal content and total metal content shall be analyzed as specified in 4.6.4.1. Calculation of individual hazardous metal contents can be based on either the manufacturer's testing of batches or the supplier's data for raw materials used in the product. When specified (see 6.2), a formulation value shall be provided that will not be exceeded when the coating system is tested in accordance with this paragraph.

4.6.4.1 Tantalum and tungsten content. The tantalum and tungsten content of the cured coating shall be determined using any appropriate spectroscopy test method. The test shall be conducted in accordance with the instrument manufacturer's directions. Data supporting the test method choice and analytical accuracy shall be established. Results shall conform to the requirements of 3.5.3.

4.6.5 Yellowness. The yellowness index of test paint shall be determined in accordance with ASTM E313. One test panel shall be exposed to normal laboratory conditions of lighting, temperature, and humidity. The second test panel shall be wrapped in light impervious paper and stored at ambient laboratory conditions in an enclosed space away from light for 12 months. After aging, the yellowness index shall be determined. Results shall conform to the requirements of 3.5.6.

4.6.6 Mixing properties. Test paint shall be thoroughly mixed by hand in accordance with the manufacturer's instructions. A paint shaker shall not be used. An amount equal to 130 grams (4.586 ounces) of the material shall be placed in a 225-gram (7.937-ounce) glass jar and not be agitated or disturbed for 2 hours. Results shall conform to the requirements of 3.6.2.

4.6.7 Spraying properties. The coating shall be mixed in accordance with the manufacturer's instructions. The coating shall be sprayed on a test panel to a dry film thickness between 0.051 and 0.076 millimeter (2 to 3 mils) and observed for spraying properties in accordance with method 4331.2 of FED-STD-141. Results shall conform to the requirements of 3.6.3.

4.6.8 Brushing properties. The coating shall be applied to a 150- by 300-millimeter Leneta chart (or equivalent) using a 75-millimeter (3-inch) brush and observe for brushing properties in accordance with method 4321.3 of FED-STD-141. Results shall conform to the requirements of 3.6.4.

4.6.9 Flexibility. Using a 3.2-millimeter (0.125-inch) mandrel, flexibility shall be determined on two test panels in accordance with ASTM D522/D522M. Results shall conform to the requirements of 3.6.5.

4.6.10 Knife test. Using a flat portion of one panel from the flexibility test, the knife test shall be performed in accordance with ASTM D6677. Any commercial sharp pocket, razor, or utility knife may be used for the test. Results shall conform to the requirements of 3.6.6.1.

4.6.11 Water resistance (does not apply to grade A and B water-based coatings). Using primed test panels, the coating shall be sprayed to the wet film thickness specified in ASTM F718 and allowed to air dry for 48 hours. After the final coat of the system has been applied, the coating shall be air dried 30 days. All exposed surfaces shall be coated. The panel shall be laid flat, and sponges wet with distilled water shall be laid on the panel in a manner that covers the full face of the test panel. The sponges shall be kept wet for a period of 7 days at ambient laboratory conditions. The sponges may be covered, but such covering shall not restrict all evaporation. Evaporation will leach water-soluble materials from the paint into the sponge. Upon removal of the sponge(s), the panel shall be observed for ASTM D1308 effects after 2 and 24 hours. Results shall conform to the requirements of 3.6.7.1.

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4.6.12 Hydrocarbon fluid resistance (does not apply to grade A and B water-based coatings). Using primed test panels, the coating shall be sprayed to a wet film thickness specified in ASTM F718 and allowed to air dry for 48 hours. After the final coat of the system has been applied, the coating shall be air dried for 30 days. All exposed surfaces shall be coated. The test panel shall be laid flat, and sponges wet with commercial kerosene or jet fuel shall be laid on the panel in a manner that covers the full face of the test panel. The sponges shall be kept wet for a period of 24 hours at ambient laboratory conditions. The sponges may be covered to restrict evaporation. On removal of the sponge(s), the panel shall be observed for ASTM D1308 effects. Results shall conform to the requirements of 3.6.7.2.

4.6.13 Salt spray resistance (does not apply to application C, D, and E coatings). Using primed steel test panels, the coating shall be sprayed to a wet film thickness specified in ASTM F718 and allowed to air dry for 48 hours. After the final coat of the system has been applied, the coating shall be air dried 30 days. All exposed surfaces shall be coated. The unscored panels shall be exposed to 5 percent salt spray for 14 days in accordance with ASTM B117. Upon removal, the panels shall be washed gently in warm running water (not more than 38 °C [100 °F]) until free from any visible salt deposits and examined immediately in accordance with ASTM D610. Results shall conform to the requirements of 3.6.7.3.

4.6.14 Condensation blister resistance. Using primed test panels, the coating shall be sprayed to a wet film thickness specified in ASTM F718 and allowed to air dry for 48 hours. After the final coat of the system has been applied, the coating shall be air dried for 30 days. All exposed surfaces shall be coated. The test shall be conducted in accordance with ASTM D2247 for 100 hours. Evaluation will be conducted, and blisters rated in accordance with ASTM D714. Results shall conform to the requirements of 3.6.7.4.

4.6.15 Exudate formation (grade A and B water-based coatings only). Two test panels shall be prepared in accordance with 4.5.1. Immediately after spraying, the test panels shall be placed in a cool place (2 to 7 °C [35.6 to 44.6 °F]) for 24 hours. After 24 hours, the film surface shall be examined for liquid residues and tackiness. Results shall conform to the requirements of 3.6.8.

4.6.16 Washability. Washability shall be determined in accordance with ASTM D4828. Results shall conform to the requirements of 3.6.9.

4.6.17 Flash rust resistance (grade A and B, application A water-based paints only). One coat of the test paint shall be spray applied to unprimed, degreased, bare steel panels, allowed to dry overnight at ambient laboratory conditions, and inspected for rust spotting. Results shall conform to the requirements of 3.6.10.

4.6.18 Mold/mildew resistance.

4.6.18.1 Standard mold resistance. The coating shall be applied at the manufacturer's maximum wet film thickness to three panels prepared in accordance with 4.5.1 under controlled laboratory conditions. The test shall be performed in accordance with ASTM D3273. Results shall conform to the requirements of 3.6.11.1.

4.6.18.2 High mold resistance (class 1, grade A coatings only). The test shall be performed in accordance with ASTM D5590. Results shall conform to the requirements of 3.6.11.2 .

4.6.19 Freeze/thaw resistance and package stability. Three sealed, unopened containers shall be tested for freeze-thaw stability in accordance with ASTM D2243 for five cycles. Results shall conform to the requirements of 3.6.12.

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4.6.20 Recoatability. Five test panels shall be prepared by applying the paint to a nonporous substrate to the recommended film thickness. Four of the test panels shall be heated in an oven for 7 days at 38 to 43 °C (100.4 to 109.4 °F). One panel shall be aged at ambient conditions for the period recommended by the manufacturer before overcoating and then overcoated with itself to the wet film thickness specified in ASTM F718. The four oven dried panels shall be individually overcoated with the originally applied paint, a water-based coating qualified to this specification, a solvent-based paint qualified to this specification, or paint in accordance with MIL-DTL-24607. All recoated panels shall be allowed to dry for at least 48 hours at normal laboratory conditions. The dry, overcoated paint film shall be inspected for wrinkling, blistering, or lifting of the test paint film. When dry, the overcoated paint film shall be cut with a knife blade and the overcoat shall be inspected for loss of adhesion and for flaking away from the test paint film. The surface of the recoat shall be inspected for smoothness, blistering, and uncoated areas. Results shall conform to the requirements of 3.6.13.

4.6.21 Thermal properties.

4.6.21.1 Painting of test specimens. Primed test specimens shall be painted to either a 0.076-millimeter (3-mil) dry film thickness of test enamel in at least two, but not more than four, coats to achieve the thickness or to the recommended dry film thickness and number of coats specified in ASTM F718. Final coating thickness shall be the greater of these two values. If multiple coats are required to achieve the test thickness, at least 24 hours, but no more than 48 hours, shall lapse between coats. Fully coated test specimens shall be aged 14 days before testing. Application of test coatings may be by brush, roller, or spray. Unless otherwise specified in the specific test method, test specimens shall be of a material corresponding to the application for which qualification is being sought.

4.6.21.2 Radiant panel test. The radiant panel test shall be conducted in accordance with ASTM E162. Test panels shall be prepared in accordance with 4.6.21.1. Results shall conform to the requirements of 3.7.1.

4.6.21.3 Smoke density chamber test. This test shall be conducted in accordance with ASTM E662 in both flaming and nonflaming modes for smoke density. Smoke produced from ASTM E662 shall be analyzed in accordance with ASTM E800. The test sample appropriate to the application for which qualification is sought shall be prepared in accordance with 4.6.21.1 on an applicable substrate. Substrates shall be primed where appropriate. Results shall conform to the requirements of 3.7.2. Smoke shall be analyzed to demonstrate compliance with 3.7.4.

4.6.21.4 Resistance to ignition. The test shall be conducted in accordance with appendix A. Results shall conform to the requirements of 3.7.3.

4.6.22 Condensation (class 4 only). DELETED.

4.7 Toxicity. A Health Hazard Assessment (HHA) will be conducted to ensure conformance to 3.10, as required by the qualifying activity. The Navy and Marine Corps Force Health Protection Command (NMCFHPC) will evaluate the coating systems using data provided by the manufacturer/distributor to the NMCFHPC (see 3.10 and 6.6).

4.8 Prohibited materials. Prohibited materials shall be verified for conformance to 3.11 as required by the qualifying activity.

4.9 Off-gassing (class 2 only). The coating shall be evaluated for off-gassing in accordance with S9510-AB-ATM-010 chapter titled “Material Control Program” (see 3.12 and 6.7). If the Navy determines that off-gas testing is required, testing shall be conducted at a NAVSEA approved test facility (see 3.12). The Navy will review the off-gas test results and assign a usage category. Additionally, the Navy will assign a usage category if an administrative review is conducted in lieu of off-gas testing (see 3.12).

4.10 Fire testing provisions. All fire tests specified in this document shall be conducted by an independent testing laboratory that is accredited to ISO/IEC 17025 and is acceptable to NAVSEA. Accreditation shall be obtained from a recognized accreditation body such as the American Association for Laboratory Accreditation (A2LA) (www.a2la.org) or the International Code Council’s International Accreditation Services (IAS) (www.iasonline.org). The scope of accreditation shall include specific flammability and fire tests required for qualification. All other fire test provisions shall be as specified (see 6.2 and 6.8).

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4.11 Label. An example of the can label, or a photograph of the label on a can, shall be provided to NAVSEA examination to determine conformance with the requirements specified in 3.9.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. These coatings are intended for application as nonflaming coatings for steel, aluminum, and nonmetallic substrates. Wherever possible, the coating should be applied over a surface that has been cleaned and primed to promote adhesion and serviceability when used on metals. Except for trace levels, all grades are free of lead, chromium, cadmium, and asbestos (see 3.5.3). All grades have levels of VOHAPs that do not exceed applicable 40 CFR 63, Subpart II levels (see 3.5.2 and 3.9). Dry coating residues and debris are non-hazardous waste under U.S. Environmental Protection Agency (EPA) regulations in effect on the date of this specification. This specification covers high-grade, air-drying coatings based on acrylic or alkyd resin that are intended for use on primed interior surfaces aboard naval ships. Existing stock of the coating is acceptable for use until depleted; future requisitions for these coatings should refer to the current version of this specification.

6.1.1 Fire resistance. The paints qualified to this specification will neither actively support combustion nor contribute to the spread of a fire.

6.1.2 General submarine interior use. Class 2 materials are designed for use in the enclosed volume of a submarine and are compatible with the operation of the atmosphere control system (emission of volatiles and thermal degradation products).

6.1.3 Military unique properties. The paint described by this specification is military unique because of the requirements that the paint be non-burning, non-hazardous, and low-smoke producing when applied to metals. Commercial products are not designed for application to metals but are intended for use on wood or plaster board.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. Class, grade, and application (see 1.2).
c. Composition (see 3.2).
d. Color (SAE AMS-STD-595 color designation nomenclature recommended; see table I).
e. Identification characteristics (see 3.4).
f. When conformance inspection is required (see 4.4).
g. Thermal properties (see 3.7).
h. DELETED
i. Manufacturer's directions for use (ASTM F718).
j. Test panel preparation, conditioning of test panels, and application of test coatings (see 4.5.1).
k. Metals content (see 4.6.4).
l. Additional fire testing provisions (see 4.10 and 6.8).
m. Packaging (see 5.1 and 6.9).
n. When a Safety Data Sheet (SDS) is required (see 6.4).

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o. When product user information sheets are required (see 6.4.1).

p. Batch VOC certification (see 6.5).

q. Acceptance test report (see B.3).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in the Qualified Products Database (QPD) No. 24596, whether or not such products have actually been so listed by that date. The attention of the manufacturer is called to these requirements and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.4 SDS. When specified (see 6.2), contracting officers will identify those activities requiring copies of completed SDSs prepared in accordance with appendix D of 29 CFR 1910.1200. Additional required Government information is contained in appendix D of 29 CFR 1910.1200. In order to obtain the SDS, federal acquisition regulation (FAR) clause 52.223-3 will be in the contract. The contracting activity should be provided an SDS at the time of contract award. The SDS should be included with each shipment of the material covered by this specification.

6.4.1 Product user information sheets. Users of coatings procured to this specification need information on the proper use, storage, and disposal of these materials. When specified (see 6.2), contracting officers will identify those activities requiring copies of completed product use sheets prepared in accordance with ASTM F718. In order to obtain these documents, an appropriate FAR clause must be in the contract.

6.5 Volatile content. When specified (see 6.2), contracting officers will require the manufacturer to provide the Government with a written VOC certification, as described in 40 CFR 63.785(a)(2), for each batch of coating product from which any quantity of the coating product is supplied to the Government under the contract. The Government reserves the right to reject as defective any coating product for which the VOC certification has not been provided.

6.6 Toxicity evaluation. The NMCFHPC requires sufficient information to permit an HHA of the product. Upon completion of the HHA, a copy will be provided by the NMCFHPC to the Government for evaluation. The HHA process is described on the NMCFHPC's website, https://www.med.navy.mil/Navy-and-Marine-Corps-Force-Health-Protection-Command/Environmental-Health/Industrial-Hygiene/Acquisition-Technical-Support/Health-Hazard-Assessments-HHA/.

6.7 Material certification. Materials to be installed in submarines are to be controlled to prevent off-gassing, which contaminates the submarine's atmosphere and can result in health hazards to personnel or deleterious effects on machinery. These controls are administered through the Submarine Material Control Program, which is described in the Nuclear Powered Submarine Atmosphere Control Manual, S9510-AB-ATM-010 chapter titled "Material Control Program." Under the Submarine Material Control Program, all materials considered for use on submarines require certification and assignment of a usage category. Under the certification process, candidate materials are selected by Navy activities or contractors, and a request for certification is submitted to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil. The certification request is accompanied by detailed information, including descriptions of the material, method of application, usage, and storage. A chemical analysis is conducted, which is accomplished through off-gas testing. If off-gas test is required, it must be conducted in a Government approved laboratory. Information pertaining to this test requirement may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil. Based on the chemical analysis results, a usage category is assigned to the material defining whether, and to what extent, the material may be used on submarines.

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6.8 Additional fire testing provisions. NAVSEA reserves the right to witness the tests and perform any of the tests set forth herein where such testing is deemed necessary to assure compliance to prescribed requirements of the qualification tests. A2LA (www.a2la.org) or International Code Council's IAS (www.iasonline.org) maintain a full list of the approved laboratories that are accredited to ISO/IEC 17025 and provide details on how to obtain and maintain accreditation (see 6.2).

6.9 Suggested packaging. Recommended packaging requirements are provided in table V (see 6.2).

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TABLE V. Packaging.

Packaging	Recommended requirements for direct Government acquisitions
Containers	1. The components should be furnished in cans capable of holding 0.47 liter (1 pint), 0.945 liter (1 quart), 3.78 liters (1 gallon), and 18.9 liters (5 gallons). In addition, coatings may also be furnished in cartridges, packs, and aerosol cans as defined in the acquisition document.
	2. Multiple friction plug containers should be in accordance with PPP-C-96, type V, class 2. Interior coatings should be as specified therein. Wire handles, as specified therein, should be provided for the 1-gallon container. Closure of the properly filled and sealed cans should be as specified in the appendix thereto.
	3. All shipping containers should comply with the requirements of the Uniform Freight Classifications (UFC), the National Motor Freight Classification (NMFC), and the applicable requirements of 49 CFR and all United Nations (UN) Model Regulations.
	4. Unit of procurement: the coatings covered by this specification should be purchased by volume. The unit of procurement should be in multiples of 1 liter or 1 U.S. liquid gallon at 16 to 27 °C (60 to 80 °F). Volumes less than 1 liter or 1 U.S. liquid gallon furnished in cartridges, packs, and aerosol cans should be purchased as defined in the acquisition document.
Packaging to ASTM D3951	1. Packaging considered commercial in nature should conform to the requirements of ASTM D3951.
	2. All containers should comply with the requirements of the UFC, the NMFC, and the applicable requirements of 49 CFR and all UN Model Regulations.
Level of pack (Packaging should be specified as follows)	1. Overseas delivery packing. Intermediate containers of like-size coating material should be packed in close-fitting wood boxes in accordance with ASTM D6251/D6251M, class 2, or ASTM D6880/D6880M, class 2. Box closure and strapping should be as specified in the applicable box specification or the appendix thereto, except that strapping should be in accordance with ASTM D3953, type 1, finish B.
	2. Domestic delivery packing. Domestic delivery packing should be as for overseas delivery, except that boxes should be class 1 and the strapping should be finish A or B.
Palletization	Intermediate containers should be palletized in accordance with MIL-STD-147.
Packing for acquisitions involving direct delivery to Navy ships or installations	Treated lumber and plywood. All lumber and plywood, including laminated veneer materials, used in shipping container and pallet construction, member, blocking, bracing, and reinforcing must be fire-retardant treated material in accordance with MIL-L-19140 as follows: a. General use, weather resistant: MIL-L-19140, type II, category I. b. General use, non-weather resistant: MIL-L-19140, type I, category I.
Unit	The coatings covered by this specification should be packed and packaged in cans, cartridges, packs, and aerosol cans as defined in the acquisition document.
SDS and product/procedure data sheets	A copy of the SDS and company product data/procedure sheets should be attached to the shipping document for each destination (see 6.4).

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TABLE V. Packaging – Continued.

Packaging	Recommended requirements for direct Government acquisitions
Hazardous warnings	1. Labels should be in accordance with 29 CFR Parts 1910, 1915, 1917, 1918, 1926, and 1928, as well as 49 CFR.
	2. All individual containers should have the following marking: "CAUTION: This material contains volatile solvents, with probable hazardous vapors. Use with adequate ventilation. Avoid prolonged breathing of vapors or spray mists. The solvents are highly flammable; avoid open flame and smoking."
	3. Each component container, shipping container, and palletized load should be marked with the appropriate hazardous symbol.
	4. Unit containers should be marked: "This product is asbestos, lead, hexavalent chromium, and cadmium free, except for possible trace levels."
VOC	VOCs should be stated on the label using the following statement: "Contains (insert VOC) grams per liter (insert VOC in lb/gal) of volatile organic content per 40 CFR 60, Appendix A-7, method 24."
VOHAPs content	VOHAP content should be stated on the label using the following statement: "Contains (insert VOHAP content here in gram/liter and pounds/gallon) solids (nonvolatiles) per 40 CFR 63."
Shelf life	Labels should include the product's manufacturer date and expiration date and meet the requirements of MIL-STD-129.
Application instructions	Manufacturers should include an ASTM F718 with each shipment of the material covered by this specification.

6.10 Subject term (key word) listing.

Hazardous materials

Solvent-based

Thermal properties

Volatile organic content

Volatiles

Water-based

6.11 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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RESISTANCE TO IGNITION

A.1 SCOPE

A.1.1 Scope. This procedure determines whether a paint will ignite into flames when the sole ignition source is the heated metallic substrate to which the paint adheres. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

A.1.2 Auto-ignition resistance. The test yields information on the ability of the coating to resist auto-ignition, running, and dripping when subjected to a very rapid heat rise, and information on the ability of the paint to resist spread of the flame across the test specimen, as well as the tendency of a paint to ignite secondary fires by flaming drips.

A.1.3 Test conditions. The conditions of the test have been standardized so that paints passing this test may be safely used on ships' bulkheads and overheads where nonflaming paints are required to prevent fires from spreading from one sealed compartment to the next during a severe shipboard fire.

A.2 TEST APPARATUS

A.2.1 Test apparatus. The recommended apparatus consists of a high amperage direct current (DC) generator capable of 300 to 400 amperes, electrodes for holding the test specimen, an ammeter attached across an in-line shunt, a voltmeter, and a switch capable of handling the voltages and currents involved. The wiring diagram is shown on figure A-1. The current from the DC generator is shunted directly through the test specimen to cause rapid heating. A 3/0 stranded welding cable is recommended for wiring. The test apparatus shall be arranged in a shielded cabinet or room to reduce air current effects. Air moving past the specimen shall be regulated to that caused only by convection heating of the test specimen.

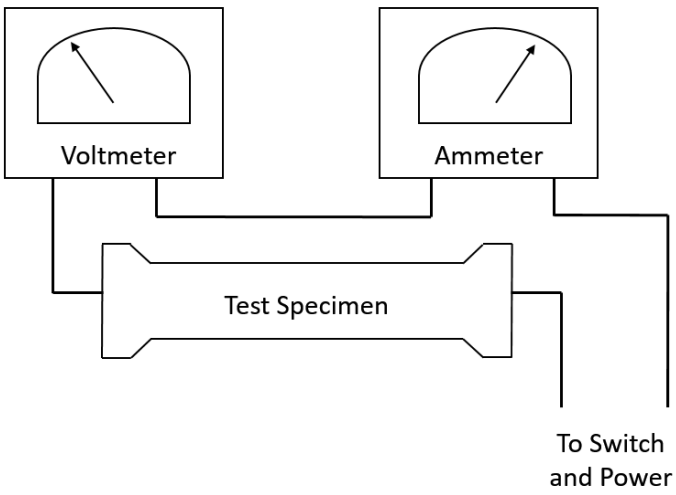


FIGURE A-1. Wiring diagram.

A.2.2 Test panels preparation. Two test specimens shall be prepared (figure A-2) by coating one specimen with MIL-DTL-24607 and one with the test nonflaming, fire-resistant coating. Each test specimen shall be coated with a 0.076- millimeters (3 mils), dry film thickness of coating, or the dry film thickness recommended by the manufacturer's data sheet, whichever is greater. If multiple coats are required to achieve the test dry film thickness, allow at least 24 hours, but not more than 48 hours of drying time at 16 to 27 °C (60 to 80 °F) and 50±10 percent relative humidity between coats. Fully coated test specimens shall be aged 14 days before being tested

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A.3 PROCEDURE

A.3.1 Test procedure. Using a test specimen (figure A-2 provides recommended specimen) made of mild steel, the apparatus shall be calibrated to give a temperature rise in the test specimen of 1,000 °C (1,832 °F) within 60 seconds of the start of the test. A timing device accurate to 1 second shall be used to determine test duration. The test specimen shown on figure A-2 is recommended and has been used by most testing organizations. The only required dimensions are that the center narrow bar shall be not less than 150 millimeters (6 inches) in length, 25 millimeters (1 inch) in width, and 1.6 millimeters (0.063 inch) in thickness. A thickness of 1.6 millimeter or greater will allow the full time of the test without the test bar burning through.

A.3.1.1 Calibration of temperature. Calibration of the 1,000 °C (1,832 °F) in 60 seconds time-temperature curves for the particular test specimens and power source shall be conducted. The time-temperature curve is determined by the cross section and resistivity of the actual test specimens used. Temperature-indicating crayons, temperature-indicating paints, or optical pyrometers are recommended. Thermocouples are not recommended due to the current flowing in the test specimen.

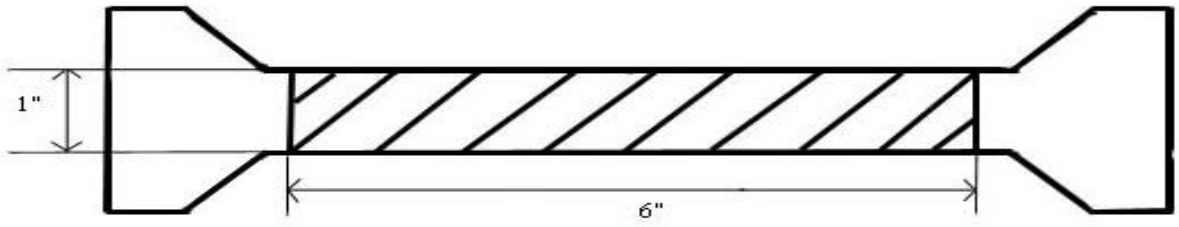


FIGURE A-2. Test specimen.

A.3.2 Data acquisition. The test specimen shall be observed and data recorded for:

a. Flaming (ignition) of the paint before reaching 1,000 °C (1,832 °F).
b. Fumes or smoke from the paint and the time at which fumes/smoke began.
c. Volume of smoke (light, moderate, or heavy).
d. Other characteristics, such as sloughing, dripping, softening, or color changes.
e. Color pictures and video shall be taken during and after the test, particularly at the time of ignition and at the point of maximum involvement.

A.3.3 Data analysis. The performance of the test nonflaming, fire-resistant coating shall be compared to the performance of MIL-DTL-24607 tested at a 0.076-millimeter (3 mils) thickness when prepared and tested under identical conditions. Test results shall be in accordance with the requirements of 3.7.3.

A.4 SUMMARY

A.4.1 Summary. This test procedure provides a relatively simple, straightforward, and reproducible method to determine the ignition potential of materials used to paint the interior surfaces of a compartment. In addition, important data on the reaction of the paint to heating of the substrate is determined. The fact that the data is of a “go” or “no-go” form makes the error of operation relatively small.

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APPENDIX B

NAVY FORMULA 25A

B.1 SCOPE

B.1.1 Scope. This appendix defines the formulation requirements for Navy formula 25A paints, including chemical composition and test requirements. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

B.2 NAVY FORMULA 25A

Table B-I contains a list of materials included in Navy formula 25A.

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TABLE B-I. Materials in Navy formula 25A.

Raw Material	Pounds	Gallons
Chlorinated emulsion 1/	699.86	71.78
Titanium dioxide 2/	199.96	5.93
Water	67.78	8.16
Aluminum trihydrate 3/	49.99	2.48
Propylene glycol 4/	30.21	3.50
Coalescing solvent 5/	21.52	2.75
Fungicide 6/	12.75	1.48
Plasticizer 7/	10.00	0.94
Thickener 8/	9.12	1.00
Dispersant 9/	8.73	0.95
Defoamer 10/	5.59	0.72
Bactericide 11/	1.27	0.13
Yellow oxide tint 12/	1.16	0.09
TOTALS	1,117.94	99.91

NOTES:

1/ Polyvinylidene chloride acrylic emulsion, 55 percent solids in water. Polidene 777, Borchers Americas, Inc 811 Sharon Drive Westlake, Ohio 44145 or SERFENE 777 produced by Owensboro Specialty Polymers Group in Owensboro, KY 42303, or equal.

2/ The titanium dioxide conforms to ASTM D476, types II or III. This formulation uses Tioxide TR-92. Huntsman Corporation, The Woodlands, TX 77380, or equal.

3/ Aluminum hydroxide with a mean particle diameter of 3-4 microns. This formulation uses SB-632 produced by Huber Engineered Materials, J.M. Huber Corporation, Edison, NJ 08837, or equal.

4/ Propylene Glycol, CAS # 57-55-6. This formulation uses the product from Lyondell-Basell Equistar Chemicals, Houston, TX 77010 or Dow Chemical Company, Midland, MI 48640, or equal.

5/ Texanol. 2,2,4-Trimethyl-1, 3-Pentanediol. CAS # 25265-77-4. This formulation uses the product from Eastman Chemical Company, Kingsport, TN 37660, or equal.

6/ Polyphase P-20T. This formulation uses the product from Troy Chemical, Newark, NJ 07105, or equal.

7/ A mixture of cyclic phosphate esters. This formulation uses Amgard CT produced by Solvay USA, Houston, TX 77098, or equal.

8/ A blend of polyurethane resin, propylene glycol, and water. This formulation uses RM-8 produced by the Dow Chemical Company, Midland, MI 48640, or equal.

9/ Tamol 851. This formulation uses the product from Dow Chemical Company, Midland, MI 48640, or equal.

10/ Foamaster MO-NDW. This formulation uses the product from BASF, Florham Park, NJ 07932, or equal.

11/ Troysan 174. 2([Hydroxymethyl] amino) ethanol. CAS # 34375-28-5. This formulation uses the product from Troy Chemical, Newark, NJ 07105, or equal.

12/ UCD 1625Q. Chromaflo Technologies Corp, Ashtabula, Ohio 44004, or equal.

B.3 ACCEPTANCE TEST REPORT

B.3.1 Acceptance test report. When specified by the qualifying activity (see 6.2), an acceptance test report shall be provided to verify the table II identification characteristics submitted.

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TABLE B-II. Test procedures.

Item	Requirement
Color	Color shall match the designated SAE AMS-STD-595 color standard. Delta E-1.0 maximum.
Total solids	table II
Vehicle solids	table II
Mass per unit volume	table II
Flash point (°C [°F])	37.8 °C (100 °F) minimum
Specular gloss (60 degrees)	45-60
Drying time – set to touch – dry hard	3 hours maximum 6 hours maximum
Hiding power (contrast ratio)	table II
Condition in container	Coating shall be free of grit, seeds, skins, lumps, putrefaction, rancidity, or livering, and shall show no more surface float, pigment settling, or caking than can be readily reincorporated to a smooth, uniform state. The container shall be free of corrosion.
Mixing properties	When mixed, the coating shall result in a smooth uniform mixture. After aging, the coating shall show no heavy settling, caking, or gelation.
Spraying properties	The coating shall spray satisfactorily with no running, sagging, or streaking. The dry film shall show no dusting, mottling, or color separation and shall present a smooth finish.
Brushing properties	The coating shall brush satisfactorily and shall dry to a smooth uniform film, free from seeds, runs, and sags.
Other test data	
Hegman grind	5 minimum
VOC, grams/liter (pounds/gallon)	175 (1.45) maximum
Leneta sag resistance (mils)	4 minimum
Viscosity (krebs units)	105 KU maximum

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CONCLUDING MATERIAL

Preparing activity:
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